



Financial Modeling Certification (FMC™) Program

Boot Camp Packet

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Introduction

Whether you're looking for a financial analyst position in investment banking, private equity, equity research or other finance-related fields, Adventis' Financial Modeling Certification (FMC™) Program gives undergraduate students the analytical and financial modeling skills necessary to differentiate themselves in today's competitive hiring landscape.

The FMC™ Program consists of two certifications that require trainees to pass a rigorous and thorough set of requirements that demonstrate they have strong analytical capabilities and an understanding of key concepts in financial modeling, accounting, corporate finance, intrinsic and relative valuation and leveraged buyout transactions. Our courses consist of trainees building best-of-class, three-statement financial models and other analyses entirely from "scratch", while selectively using lecture-based instruction to explain concepts along the way.

When employers see a resume with a FMC™ Level I Certification or FMC™ Level II Certification, they know that the candidate has the analytical abilities to be a great analyst. Trainees are taught and certified by current and former investment banking, private equity, and equity research professionals. They have participated in two full-day, instructor-led courses and practiced their financial modeling skills outside of the courses, typically for a total of 30-45 hours. Certified trainees have been vetted as having mastered the basic financial modeling and valuation skills required to excel in demanding and fast paced work environments. The process of getting certified helps not only to differentiate students, but it is extremely similar to the preparation process students must go through in order to be ready for interviews.

Please use this Boot Camp Packet to prepare for our courses and certification exams.

We hope you enjoy your participation in the FMC™ Program!

Regards,

A handwritten signature in black ink, appearing to read 'D. Lewicki', with a stylized, cursive script.

David Lewicki
Founder and CEO

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1. Instructions for the Self-Paced Exercise

One of the unique characteristics of the FMC™ Program is the Self-Paced Exercise that most students complete within 5 minutes in advance of our instructor-led courses. This exercise is designed to teach trainees best-of-class formatting techniques and Microsoft Excel keystrokes. This approach enables trainees to focus almost exclusively on learning concepts related to financial modeling during the instructor-led components of the FMC™ Program.

Please see the emails you receive after you enroll for more details on how to best complete the Self-Paced Exercise. The objective of this exercise is for you to internalize the formatting techniques and Microsoft Excel skills that will be utilized in class. You should expect to spend up to 2-3 hours in order to complete the Self-Paced Exercise within 5 minutes.

2. Adventis' Financial Modeling Macros

Adventis' Financial Modeling Macros are contained in an Excel file. The Macros will make you more efficient and enhance your learning curve in regards to building financial models. Gaining a familiarity with the Macros will help you well beyond your participation in the FMC™ Program. You must have the Macros open while working on the Self-Paced Exercise and attending courses in the FMC™ Program. Please see below for a depiction of the Macros for PCs. At the bottom, there are instructions. While we strongly recommend using a PC, we also offer Macros for Macs in our portal.

Expires on December 31, 2021



Financial Modeling Macros (for PCs)

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Macro	Key Stroke	Result
Font Color	Ctrl + Shift + L	Toggle: Text Color Text Color Text Color Text Color
Digits	Ctrl + Shift + D	Toggle: \$12.3 12.3 1,234.0% 12.3x
Decimals	Ctrl + Alt + Left / Right Arrow Keys	# of Decimals: ...← 0.0000 ← 0.000 ← 0.00 ← 0.0 ← 0 ...→ 0.0000 → 0.000 → 0.00 → 0.0 → 0
Indent	Ctrl + Shift + I	Toggle: Left Indent = 1 Left Indent = 0 Left Indent = 1
Cell Border	Ctrl + Shift + Alt + All Arrow Keys	Toggle: ← ↑ → ↓
Highlight	Ctrl + Shift + H	Toggle: No Fill Yellow Fill Orange Fill Blue Fill
Font Size	Ctrl + Alt + Up / Down Arrow Keys	Increase (up) / decrease (down) font size by 1.0
Accounting Underline	Ctrl + Shift + U	Single Accounting Underline and Center Across
Justification	Ctrl + Shift + J	Toggle: Left Align Center Right Align
Center Vertically	Ctrl + Shift + M	Center Text Vertically
Gridlines	Ctrl + Shift + G	Show / Hide Gridlines
Freeze Panes	Ctrl + Shift + F	Freeze / Unfreeze Panes Around Selected Cell
Page Settings	Ctrl + Shift + P	Set page settings on landscape, 0.5 margins, page # on bottom center, 100% zoom
Trace Dependents	Ctrl + Shift + X	Trace cells that provide data to a formula
If (Isnumber)	Ctrl + Shift + E	=IF(ISNUMBER(formula),formula,0)

Instructions:

1. Make sure you save down the Macros to your hard drive of the computer you are using
2. Enable the Macros by clicking the File Menu (Office Backstage) and selecting Excel Options from the left sidebar. In options, select Trust Center from the left sidebar and click the Trust Center Settings button on the main window. Now in Trust Center Settings dialog window, select Macro Settings from the left sidebar, choose Enable All Macros option and hit OK.
3. Disabled hotkeys by right clicking on your desktop screen and selecting Graphic Options. Then select Hotkeys and then Disable.
4. Remove any add-ins by clicking the File Menu (Office Backstage) and selecting Excel Options from the left sidebar. In options, select Add-Ins from the left sidebar and click the Go button on the main window. Now in Add-Ins window, uncheck any boxes that are checked and hit OK.
5. After all appropriate changes have been made, close down Excel and re-open the Macros and other relevant Excel files.

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3. Keyboard Shortcuts

Learning how to use your computer with limited use of the mouse is critical for efficiency in financial modeling, as well as using other applications on your computer. Please see below for 40 critical keyboard shortcuts you will use in the Self-Paced Exercise and during courses in the FMC™ Program (numbers 30-40 refer to Adventis' Financial Modeling Macros). Please note that, for most of these keystrokes, if you hit the Alt button you will see a number of letters in the ribbon that you can then use to find shortcuts.

Keyboard Shortcuts and Adventis' Financial Modeling Macros			
No.	Action	PC Keystroke	Mac Keystroke
Basic Keystrokes			
1	Open selected folder	Enter	No shortcut
2	Go back to the previous folder	Backspace	No shortcut
3	Switch to the next program	Alt + Tab	Alt + `
4	See the contents of a cell / change filename	F2	Same
5	Opens item	Alt + underlined letter in menu	No shortcut
6	Start a formula	= or +	Same
7	Copy / cut / paste	Ctrl + C / X / V	Cmd + C / X / V
8	Bold / underline / italic	Ctrl + B / U / I	Cmd + B / U / I
9	Save / save as	Ctrl + S / Alt + F + A	Cmd + S
10	Print	Ctrl + P	Cmd + P
Intermediate Keystrokes			
11	Moves cursor to cell A1 (or top left of frozen pane)	Ctrl + Home	Fn + Ctrl + Left Arrow Key
12	Move down / up one screen on the same worksheet	Page Up / Down	Fn + Up / Down Arrow Keys
13	Move cursor to end of data region	Ctrl + Arrow Keys	Cmd + Arrow Keys
14	Move cursor to end of and highlight data region	Ctrl + Shift + Arrow Keys	Cmd + Shift + Arrow Keys
15	Move to prior / next worksheet	Ctrl + Page Up / Page Down	Option + Left or Right Arrow Keys
16	Move cursor to highlight data region	Shift + Arrow Keys	Same
17	Cancel a cell entry	Esc	Same
18	Undo / redo the last section	Ctrl + Z / Y	Cmd + Z / Y
19	Repeat last action	F4	Fn + F4
20	Calculates all worksheets in manual calculation mode	F9	Fn + F9
Advanced Keystrokes			
21	Paste special format / formula / value	Alt + H + V + S + T / F / V	Cmd + Option + V + T / F / V
22	Change row height / column width	Alt + H + O + H / W	No shortcut
23	Select entire column / row	Ctrl / Shift + Spacebar	Ctrl + Spacebar
24	Fill down / to the right	Ctrl + D / R	Cmd + D / R
25	Insert the sum function	Alt + =	Shift + Cmd + T
26	Go to first precedent	Ctrl + [	Cmd + {
27	Go back to previous cell	F5 + Enter	Same
28	Print preview	Alt + F + W + V	No shortcut
29	Select entire worksheet	Ctrl + A	Same
Adventis' Financial Modeling Macros (Selected)			
30	Cycle font color	Ctrl + Shift + L	Ctrl + Shift + L
31	Change number formatting	Ctrl + Shift + D	Ctrl + Shift + D
32	Add or remove decimals	Ctrl + Alt + Left / Right Arrow Keys	Ctrl + Shift + M / N
33	Cycle fill color	Ctrl + Shift + H	Ctrl + Shift + H
34	Increase or decrease font size	Ctrl + Alt + Up / Down Arrow Keys	No macro
35	Add cell borders	Ctrl + Shift + Alt + Arrow Keys	No macro
36	Add accounting underline	Ctrl + Shift + U	Ctrl + Shift + U
37	Add or remove indentation	Ctrl + Shift + I	Ctrl + Shift + I
38	Toggle better left, middle and right justification	Ctrl + Shift + J	Ctrl + Shift + J
39	Add or remove gridlines	Ctrl + Shift + G	Ctrl + Shift + G
40	Adjust page settings	Ctrl + Shift + P	Ctrl + Shift + P

Additionally, see the below table for a general listing of key differences between a PC and a Mac on keystrokes when building financial models.

Action	Commentary
Control button	Generally the Command button replaces the Control button.
Add cell borders (macro)	There is no macro for this on the Macros for Mac, so you'll have to select the border manually for sum lines
Switch to the next program	Instead of Alt+Tab on a PC, this is Command + `
Add or remove decimals	The macro for this on the Macros for Mac is Ctrl + Shift + N / M (vs Ctrl + Alt + Left / Right Arrow Keys for PCs); on some Macs this macro won't work and when that happens you should go to the Home tab of the ribbon and use the Increase Decimal or Decrease Decimal button in the Number group.
Move to prior / next worksheet	Instead of Ctrl + Page Up / Page Down on a PC, this is Option + Left or Right Arrow Keys on a Mac
Go to first precedent	Instead of Ctrl + [on a PC, this is Cmmd + { on a Mac
Setting automatic calculation	In the Menu bar select "Excel", then "Preferences, then "Calculation". Set calculation options to automatic and use iterative calculations.
Repeat last action	Instead of F4, it's Fn + F4
Calculate all worksheets in manual calculation mode	Instead of F9, it's Fn + F9

4. Boot Camp Overview & Set-Up

Boot Camp Overview

Our two-day boot camps are typically held over a weekend, on Saturday and Sunday. Each day typically lasts 7-8 hours, with an hour lunch break. On the first day, we build a three-statement financial model of a real company from “scratch” and perform a variety of valuation analyses using our financial projections, culminating in the “football field” output that depicts the different valuation ranges calculated. On the second day, build a full LBO analysis from “scratch” of the same company and a three-statement financial model of a different company.

Day One	▪ Financial Modeling Course: Build a 3-statement financial model of a real company ▪ Valuation Module: Build public & M&A comparables, a WACC analysis, a DCF analysis, and a “football field” analysis of a real company	8 Hours
Day Two	▪ Leveraged Buyout Module: Build a leveraged buyout analysis of a real company ▪ Financial Modeling Course: Build a 3-statement financial modeling of another company	7 Hours

Boot Camp Set-Up

When you enrolled in our boot camp, you received an email with attachments for both day one and day two of the boot camp. The email for each day includes selected pages of the 10-K from the company we will be building a model for, an Excel file with inputs for that company and an Excel file that you will use to build the financial model. Upon arriving at the boot camp each morning, please pull up these files on your computer and save them down on your local hard drive. Additionally, please save your Macros on your local hard drive as well. Please also make sure that in Excel Options: Formulas, Workbook Calculation is on Automatic and Enable Iterative Calculations is checked.

5. **Adventis' 10-Step Financial Modeling Method**

You will follow Adventis' 10-Step Financial Modeling Method during the Financial Modeling Course. You should also follow this method while you're practicing for our FMC™ Level I Certification.

Step	Category	Action
1	Income Statement	Paste in historical income statement figures
2	Income Statement	Project out income statement figures; leave interest expense blank for now
3	Balance Sheet	Paste in historical balance sheet figures
4	Balance Sheet	Flat line balance sheet figures; the balance sheet should be balanced from here out
5	Cash Flow Statement	Format the cash flow statement
6	PP&E Schedule	Build out the PP&E schedule and link into the financial statements
7	Working Capital Schedule	Build out the working capital schedule and link into the financial statements
8	Equity and Related Schedules	Build out the equity, common stock, stock purchase and dividend schedules and link into the financial statements
9	Debt Schedule	Build out the debt schedule and link into the financial statements
10	Interest Expense Schedule	Build out the interest expense schedule and link into the financial statements

6. Steps for Valuation Course

Below are some basic steps you will use in the Valuation Course to build a discounted cash flow analysis.

Public Comparables Analysis

Step	Category	Action
1	Build out the full comparables set, then fill in for the company we are analyzing	Fill in the calculations from left to right for the comparables set, then do the same for the company we are analyzing

Acquisition Comparables Analysis

Step	Category	Action
1	Build out the full comparables set	Fill in the calculations from left to right for the comparables set

WACC Analysis

Step	Category	Action
1	Link in inputs	Link in the corporate tax rate, common shares, share price, and market value of debt
2	Calculate all other metrics	Calculate the cost of debt, cost of equity, and WACC

Discounted Cash Flow Analysis

Step	Category	Action
1	Link in financial statements and establish unlevered free cash flow	Link in the appropriate line items from the income statement and cash flow statement
2	Calculate present value of projection period	Determine the present value for each year in the projection period and sum all the years
3	Determine terminal value using the perpetuity method	Determine the present value of the terminal value using the perpetuity method and sum with projection period to calculate total valuation
4	Determine terminal value using the exit multiple method	Determine the present value of the terminal value using the exit multiple method and sum with projection period to calculate total valuation

Football Field Analysis

Step	Category	Action
1	Fill in the comparable companies analysis	Link in target metrics, net debt, and shares outstanding, then calculate enterprise value, equity value, and stock price
2	Fill in the acquisition comparables analysis	Link in target metrics, net debt, and shares outstanding, then calculate enterprise value, equity value, and stock price
3	Fill in the DCF analysis	Link in enterprise value, net debt, and shares outstanding, then calculate equity value and stock price
4	Fill in the current valuation for the company we're analyzing	Hard code stock price, link in net debt and shares outstanding, then calculate enterprise value and equity value

7. Steps for Leveraged Buyout Course

Below are some basic steps you will use in the Leveraged Buyout Course to build leveraged buyout analysis.

Step	Category	Action
1	Link in financial statements	Link in appropriate line items from the income statement and cash flow statement
2	Transaction value	Determine offer price, purchase price and transaction value
3	Sources and uses	Put in assumptions for deal fees and beginning cash balance and determine appropriate debt levels
4	Debt schedule	Set up debt pay down schedule
5	Cash schedule	Set up cash schedule
6	Interest expense schedule	Set up interest expense schedule
7	Investor returns	Calculate investor returns
8	Credit metrics	Calculate credit metrics

8. Overview of Certifications

Earning our certifications requires rigorous and meticulous training that differentiates undergraduate students and professionals. We award our certifications to those that exhibit the knowledge of skills taught in our courses by passing our exams – not for simply attending our courses. Those who earn the certifications can build best-of-class, three-statement, fully circular financial models completely from “scratch” and demonstrate superior knowledge of the concepts behind financial modeling and valuation.

Levels of Certification

The FMC™ Program consists of two levels of certification. Each level requires trainees to pass a rigorous and thorough set of requirements that demonstrate strong analytical capabilities and an understanding of key concepts.

- **FMC™ Level I Certification:** Identifies trainees who have learned best-of-class financial modeling skills and internalized key financial modeling, accounting and corporate finance concepts; trainees must build an industry-leading, three-statement financial model from scratch in less than 90 minutes and score at least 90% on a 30-question multiple-choice exam.
- **FMC™ Level II Certification:** Identifies trainees who have internalized key valuation and leveraged buyout concepts, in addition to having earned our FMC™ Level I Certification. This certification requires trainees to score at least 90% on a 25-question multiple-choice exam.

	FMC™ Level I Certification	FMC™ Level II Certification
Requirements	▪ Participation in two days of instruction	▪ FMC™ Level I Certification
Skills & Concepts	▪ Excel keystrokes ▪ Basic financial modeling skills	▪ Valuation ▪ Leveraged buyout transactions
Overview of Exams	▪ Financial modeling exam (90 minutes) ▪ Multiple-choice exam (90%)	▪ Multiple-choice exam (90%)

9. Overview of the Financial Statements

Financial statements are a set of three integrated reports: income statement, balance sheet and cash flow statement. The financial statements communicate the financial condition, results of operations, and various other activities of an organization. Financial statements provide a considerable amount of information to various stakeholders that enable them to make decisions. Some examples of how different stakeholders use financial statements are as follows:

Group	Example of How Financial Data is Used
Board of Directors	Hold management accountable and make board-level decisions about corporate strategy
Company Management	Measure performance and make strategic, operating and financial decisions
Creditors	Measure creditworthiness, liquidity and bankruptcy risk
Investors	Make decisions on buying or selling equity investments
Acquirers	Determine valuation and make investment decisions
Regulators	Determine whether the company is operating according to regulations and the law

Income Statement

The income statement presents the results of operations (e.g. profitability) over a period of time, typically monthly, quarterly and/or annually. The purpose of the income statement is to show stakeholders whether the company made or lost money during the period being reported. It indicates how revenues are transformed into net income. It displays the revenues recognized for a specific period of time and the expenses charged against those revenues. Please see below for a description of some common income statement items and a depiction of what a standard income statement might look like over a 5-year period.

- **Revenue (or Sales)** is the amount charged for the delivery of goods or services
- **Cost of Sales (or Cost of Goods Sold)** is the direct cost of producing revenue (raw materials, direct wages, etc.)
- **Gross Profit** is calculated as revenue less cost of sales and indicates how efficiently labor and supplies are used in the production process
- **Operating Expenses** are all other expenses required to run the business (management salaries, marketing, travel, etc.)
- **Operating Income (or EBIT)** is calculated as revenue less cost of sales and operating expenses and indicates a company's earning power from ongoing operations
- **Non-Operating Expenses** are expenses or income not related to the regular business of the company (interest expense, restructuring expenses, etc.)
- **Corporate Taxes** are local and federal income taxes the company incurs
- **Net Income (or Net Earnings)** is calculated as revenue less all expenses of the company and indicates the increase in shareholders' value resulting from operations

Sample Income Statement			
	2016A	2017A	2018A
Revenue	\$415	\$430	\$470
% Growth	3.8%	3.6%	9.3%
Cost of Sales	260	275	290
Gross Profit	\$155	\$155	\$180
% Margin	37.3%	36.0%	38.3%
Operating Expenses	52	53	58
Operating Income	\$103	\$102	\$122
Non-Operating Expenses	12	11	10
Income Before Tax	\$91	\$91	\$112
Corporate Taxes	10	10	18
Net Income	\$81	\$81	\$94
% Margin	19.5%	18.8%	20.0%

Balance Sheet

The balance sheet shows an organization's financial position at a particular point in time. It discloses the resources an organization controls (assets) and the claims on those resources (liabilities and equity). Said in a different way, assets are what a company owns, liabilities are what a company owes and equity represents any remaining claims of shareholders against a company.

The basic accounting equation (assets = liabilities and equity) represents the relationship between the assets, liabilities and shareholder's equity of a company. It is the foundation for the double-entry bookkeeping system, in which there is a credit for every debit. Such an arrangement ensures that total assets are always equal to total liabilities plus total shareholders' equity. Furthermore, the accrual accounting method measures the performance of a company regardless of when cash transactions occur. This method of accounting is the standard accounting practice for companies. Please see below for a description of some common balance sheet items and a depiction of what a standard balance sheet might look like over a 5-year period.

- **Cash** is current assets comprising currency or currency equivalents that can be accessed immediately
- **Accounts Receivable** is the amount owed to an organization from the sale of its products or services
- **Fixed Assets** is the value of assets and property that cannot easily be converted to cash and has a useful life of greater than 1 year. An example of Fixed Assets is Property, Plant & Equipment.
- **Accounts Payable** is the amount owed to an organization's vendors
- **Debt** is the amount of obligations owed to creditors
- **Equity** is cumulative shareholder investment plus cumulative net income

Sample Balance Sheet				
		2016A	2017A	2018A
Assets of a Company	Cash	\$227	\$284	\$350
	Accounts Receivable	70	72	80
	Total Current Assets	\$297	\$356	\$430
	Fixed Assets	103	107	111
	Total Assets	\$400	\$463	\$541
Claims on those Assets	Accounts Payable	\$53	\$55	\$60
	Short-Term Debt	25	25	25
	Total Current Liabilities	\$78	\$80	\$85
	Long-Term Debt	80	75	70
	Total Liabilities	\$158	\$155	\$155
	Shareholders' Equity	242	308	386
	Total Liabilities and Equity	\$400	\$463	\$541

Balance Sheet (Cont'd)

Working Capital

While there isn't an account on the balance sheet called working capital, it is an important financial metric derived from balance sheet accounts. Working capital is a measure of a company's efficiency and its short-term financial health. Working capital is calculated as non-cash current assets less non-debt current liabilities. If a company's non-cash current assets do not exceed non-debt current liabilities, then it may run into challenges repaying creditors and suppliers in the short run. While having positive or negative working capital isn't necessarily a good or bad thing, a positive or negative balance determines whether working capital is a source or use of cash.

- **Non-Cash Current Assets** represent all assets besides cash that are expected to be converted into cash within one year. These assets appear on the company's balance sheet and include accounts receivable, inventory, prepaid expenses, and other assets.
- **Non-Debt Current Liabilities** represent all obligations besides short-term debt that are due within one year. Current liabilities appear on the company's balance sheet and include accounts payable, accrued liabilities, and other obligations.

Debt vs. Equity

Debt is generally a less expensive form of capital, as it is less risky. Debt owners typically have priority claims on a company's assets if the company goes bankrupt. Equity is more expensive, as equity holders are not guaranteed to get their investment back if the company goes bankrupt. Therefore, investors require a higher rate of return to mitigate their risk.

Net Debt

While there isn't an account on the balance sheet called net debt, it is an important financial metric derived from balance sheet accounts. Net debt is simply total debt less cash. If cash were used to pay down debt, net debt would be the resulting amount. Net debt is primarily used in credit analysis, as creditors assume the company's cash balance could be applied to debt repayment in the event of a liquidity crunch or bankruptcy.

Cash Flow Statement

The cash flow statement shows how much cash is generated or lost during a period of time. It reconciles net income to change in cash. It shows how changes in the balance sheet accounts and net income affect cash and breaks down cash into operating, investing and financing activities. As an analytical tool, the statement of cash flows is useful in determining the short-term viability of a company, particular its ability to pay its bills. It also reflects a company's liquidity. Please see below for a description of some common cash flow statement items and a depiction of what a standard cash flow statement might look like over a 3-year period.

- **Cash from Operating Activities** is the amount of cash generated by an organization's normal business operations. The most common line items included in cash from operating activities are net earnings, depreciation and amortization, and change in working capital accounts.
- **Cash from Investing Activities** is cash flow related to the acquisition and disposal of an organization's long-term investments, including property, plant and equipment and mergers and acquisitions. The most common line items included in cash from investing activities are capital expenditures and acquisitions.
- **Cash from Financing Activities** is cash flow between an organization and its owners and creditors. The most common line items included in cash from financing activities are debt and equity issuances and repayments, dividends, and share repurchases.
- **Beginning Cash Balance** is the ending cash balance for the previous period of time (for example, the beginning balance for FY 2018 is the ending balance for FY 2017).
- **Change in Cash** is sum of the Cash from Operating, Investing, and Financing Activities.
- **Ending Cash Balance** is the resulting cash balance for the selected period of time and should be reflected on the balance sheet.

Sample Cash Flow Statement			
	2016A	2017A	2018A
Net Earnings	\$81	\$81	\$94
Depreciation	11	11	11
Change in Working Capital	(4)	--	(3)
Cash from Operating Activities	\$88	\$92	\$102
Capital Expenditures	(15)	(15)	(15)
Cash from Investing Activities	(\$15)	(\$15)	(\$15)
Change in Debt	(\$5)	(\$5)	(\$5)
Share Repurchases	(8)	(9)	(10)
Dividends	(6)	(6)	(6)
Cash from Financing Activities	(\$19)	(\$20)	(\$21)
Change in Cash	\$54	\$57	\$66
Beginning Cash	\$173	\$227	\$284
Ending Cash	227	284	350

Cash Flow Statement (Cont'd)

Depreciation & Amortization

Depreciation & Amortization is a method of allocating the cost of an asset over its useful life for both accounting and tax purposes. While this is an expense on the income statement, this amount does not actually represent cash leaving the company because cash only leaves the company when the asset is initially purchased (classified as CapEx). This is why Depreciation & Amortization is a source of cash on the cash flow statement.

Capital Expenditures

Capital Expenditures ("CapEx") are funds used by a company to purchase or upgrade physical assets such as plants, property and equipment ("PP&E").

Change in Working Capital

Change in working capital consists of the impact to cash resulting from all non-cash current asset accounts and all non-debt current liability accounts. A decrease in working capital represents a source of cash and would be represented on the cash flow statement as a positive number; an increase in working capital, however, represents a use of cash and would be represented on the cash flow statement as a negative number.

Share Repurchases

A share repurchase is the reacquisition by an organization of its own stock. The organization either retires the repurchased shares or keeps them as treasury stock. When shares are repurchased, the remaining shareholders have a higher ownership percentage. Because the resulting ownership stake held by remaining shareholders is larger, so is each shareholder's portion of earnings; thus share repurchases are a form of returning capital to shareholders and one that can be instituted irregularly versus a normal dividend program.

Dividends

Dividends are a distribution of cash to current shareholders. Dividends are most often derived from a dividend per share amount as directed by an organization's board of directors. Unlike share repurchases, dividends do not affect ownership percentages and represent a pure "check" to shareholders, though dividends are often reinvested in the business. Companies with stable cash flows might choose to institute a dividend program to constantly return cash to shareholders, but dividends become "sticky" and expected, and downgrading or removing its dividend program can signal trouble for a company.

Cash Flow Statement (Cont'd)

Change in Debt

Change in debt represents any debt issuances or repayments.

EBITDA

EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortization) is one of the most commonly used metrics in business. The EBITDA of a business gives an indication of its current operational profitability. It is widely used when assessing the performance of a company, as it allows for a comparison of profitability between companies in a wide range of industries. By excluding interest, taxes, depreciation and amortization, EBITDA excludes the effects from different forms of financing, different political and tax jurisdictions, and different rules around the depreciation and amortization of assets.

Free Cash Flow

Free Cash Flow is a way of looking at a company's cash flow to see what is available for distribution to creditors and shareholders. There are different ways to calculate free cash flow, however it is most commonly defined as cash flow from operations less capital expenditures. Free cash flow is one of the more important metrics to use when valuing a company.

10. Financial Ratios

Financial ratios are useful indicators of a company's performance and financial situation, and past ratios are often the starting point for forecasting into the future. Financial ratios can be used to spot trends, analyze the impact of drivers and variables, evaluate performance and keep companies on the right track. They are calculated from information provided by the financial statements. They can be expressed as a decimal value (0.10), a percentage (1.0%), or a multiple (10.0x). Financial ratios can be used to analyze performance, make more informed decisions and compare a company's financial situation and performance to other companies. Financial ratios are typically classified according to the information they provide. They are typically classified into the following five categories:

- **Liquidity Ratios** indicate a company's ability to meet its short-term financial obligations
- **Efficiency Ratios** indicate how effectively a company utilizes its assets
- **Profitability Ratios** provide insight into the profits made by a company relative to its assets, equity or revenue
- **Credit Ratios** measure a company's ability to meet its long-term obligations, and benchmark its overall capital structure
- **Market Ratios** measure investor response to owning a company's stock

Liquidity Ratios

Liquidity ratios indicate a company's ability to meet its short-term financial obligations. They are of most interest to those extending short-term credit to a company, such as banks and other traditional lending institutions. Current ratio and cash ratio are two common liquidity ratios.

Metric	Definition	Formula	Example
Current Ratio	Indicates whether a company's short-term assets are readily available to pay off its short-term liabilities	Current Assets / Current Liabilities	A current ratio between 1.50 and 3.00 is normal
Cash Ratio	Indicates a company's ability to use cash to pay off its current liabilities	Total Cash / Current Liabilities	A cash ratio between 0.20 and 1.00 is normal

Efficiency Ratios

Efficiency turnover ratios indicate how effectively a company utilizes its assets. Days receivable and asset turnover are two of the more common efficiency ratios.

Metric	Definition	Formula	Example
Days Receivable	Average number of days an invoice is in accounts receivable before collection	$(\text{Accounts Receivable} / \text{Annual Revenue}) \times 365$	Days receivable of 60.0 means a company's invoices are on average paid down in 60 days
Asset Turnover	Amount of revenues generated per dollar of assets; measures the efficiency of a company's use of its assets in generating sales revenue	$\text{Total Revenue} / \text{Total Assets}$	An asset turnover ratio of 2.50 means that for every dollar of assets a company generated \$2.50 in revenue

Profitability Ratios

Profitability ratios provide insight into the profits made by a company relative to its assets, equity or revenue. They measure a company's ability to generate profit relative to a metric, and evaluation of different metrics can help point to outperformance vs. peers or opportunities for improvement.

Metric	Definition	Formula	Example
Gross Margin	Relative to revenue, indicates how efficiently labor and supplies are used in the production process	$\text{Gross Profit} / \text{Revenue}$	A gross margin of 35.0% means that for every \$1.00 of revenue, \$0.35 is converted into gross profit
Operating Margin	Relative to revenue, indicates a company's earning power from ongoing operations	$\text{Operating Income} / \text{Revenue}$	An operating margin of 15.0% means that for every \$1.00 of revenue, \$0.15 is converted into operating profit
Net Margin	Relative to revenue, indicates the increase in shareholders' equity from earnings	$\text{Net Income} / \text{Revenue}$	A net margin of 10.0% means that for every \$1.00 of revenue, \$0.10 is converted into net income
Return on Equity	Measures profits earned for each dollar invested in a company's equity	$\text{Net Income} / \text{Shareholders' Equity}$	A return on equity of 17.0% means that for every \$1.00 of \$0.17 of net income is produced

Credit Ratios

Credit ratios measure a company's ability to meet its long-term obligations. They are used by creditors, ratings agencies, and management to make sure companies are properly capitalized, have the ability to safely pay down debt obligations and evaluate the impact of a possible transaction or change to capital structure.

Metric	Definition	Formula	Example
Debt / EBITDA (Total Leverage)	Used to assess the probability on defaulting on debt	Total Debt / EBITDA	A debt / EBITDA ratio of 2.5x means that it will take approximately 2.5 years to pay off the debt
Net Debt / EBITDA (Net Leverage)	Used to assess the probability on defaulting on debt, taking into account a cash balance that could be used to pay down debt	Net Debt / EBITDA	A net debt / EBITDA ratio of 2.5x means that it will take approximately 2.5 years to pay off the debt
Debt-to-Equity	Indicates the relative proportion of debt and equity used to finance a company's assets	Total Debt / Total Equity	A debt-to-equity ratio of 25% means that for every \$1.00 owned by shareholders, \$0.25 is owed to creditors
EBITDA Interest Coverage Ratio	Indicates how easily a company can pay interest on outstanding debt	EBITDA / Interest	An EBITDA coverage ratio of 2.0x means that the company is producing approximately two times the cash flow needed to satisfy interest expense requirements

Market Ratios

Market ratios measure investor response to owning a stock. They help to understand how investors value a company. Market ratios evaluate the market price of a share of common stock and are an indicator of a company's ability to generate profits or build assets. It is important to be able to distinguish between and understand Enterprise Value and Equity Value in order to understand some of the market ratios listed below. These terms are also an important step in having a good understanding of valuation techniques.

Metric	Definition	Formula	Example
Enterprise Value (Firm Value or EV)	Shows the sum of all claims on a company (the entire firm's value) and is independent of capital structure; changes in capital structure do not affect EV	Equity Value + Net Debt	An enterprise value of \$150 million means that the sum of all claims (debt and equity) on a company total to \$150 million
Equity Value (aka Market Value, Market Capitalization)	Value of the equity shareholders' portion of the company, or the value of all the shares outstanding	Enterprise Value - Net Debt or Share Price x Shares Outstanding	An equity value of \$100 million means the current value attributable to shareholders is \$100 million
Price / Earnings (P/E)	Shows how much investors are willing to pay per dollar of earnings and is affected by leverage	Market Value / Net Earnings	A P/E ratio of 15.0x means that investors are willing to pay \$15.00 for every \$1.00 of net income
Earnings per Share (EPS)	Shows the amount of earnings attributable to a share of common stock	Net Earnings / Shares Outstanding	An EPS of \$2.45 means that \$2.45 was earned for every share of stock outstanding
Dividend Yield	Shows the return on a share of common stock	Dividend per Share / Market Price per Share	A dividend yield of 20% means that for every \$1.00 in stock price value, \$0.20 was distributed to shareholders as dividends
Enterprise Value / EBITDA	Measures the value of common stock in a way that enables comparison of companies across different industries; EBIT can also be used	Enterprise Value / EBITDA	An EV / EBITDA multiple of 4.5x means that for every \$1.00 of EBITDA the company is worth \$4.50
Enterprise Value / Revenue	Compares the total value of a company to its revenue	Enterprise Value / Revenue	An EV / revenue multiple of 1.5x means that for every \$1.00 of revenue the company is worth \$1.50

11. Overview of Financial Modeling

Financial modeling is a general term that means different things to different audiences. Financial modeling in the FMC™ Program refers to financial statement forecasting.

A financial model is a representation of an organization's financial path forward. Unless a financial model is being built for a start-up organization, financial modelers will rely on historical financial statements to help them forecast financial statements. The two primary objectives of financial modeling are to arm decision makers with reliable information and to communicate effectively to stakeholders. Applications of financial modeling include business valuations, scenario planning, budgeting, financial statement analysis, project finance, and mergers and acquisitions. In financial models, calculations and outputs are driven by inputs. Although other software applications exist, the vast majority of financial models are built in Microsoft Excel.

In order to be able to project the three financial statements, a basic understanding of how the three statements work together is critical. Please see below for a summary of some example transactions that impact all three financial statements:

Transaction	Impact to Financial Statements	
Positive net income	Equity increases	Cash increases
Pay down debt	Debt decreases	Cash decreases
Submit an invoice to a customer	Revenue increases	Accounts receivable increases
Receive payment for an invoice	Accounts receivable decreases	Cash increases
Pay a bill	Accounts payable decreases	Cash decreases
Purchase equipment	Fixed assets increase	Cash decreases

Financial Modeling Best Practices

Financial modeling best practices vary on a firm-by-firm and industry-by-industry basis; however, there are some core objectives to best-of-class financial modeling best practices:

- Financial models should be easily understood by all audiences
- The process of building financial models should be as efficient as possible
- Making changes to financial models should be seamless
- Models should be transferable from original author with ease
- Data should be accurate and represented in the appropriate level of detail

Furthermore, there are a number of tactical financial modeling best practices that are important to follow as you are building, iterating, and maintaining financial models. These best practices serve to aid in achieving the objectives set forth above. While the below list is not exhaustive, it includes some of the core financial modeling best practices.

Category	Method	Why?
Font Color	▪ Inputs: Blue ▪ Links to other sheets: Green ▪ Calculations & titles: Black ▪ Temporary attention grabbers: Red	Enables easy differentiation between inputs, links, and calculations
Numbers	▪ Negative numbers in parenthesis ▪ “0” is reflected as “- -”	Properly aligns all digits and is easier to read; shows the difference between 0 and 0.1
Currency	▪ Currency symbols are used in the top and bottom rows of a list of financial figures, and there are never two next to each other	Shows the user know what currency the data is in and is visually appealing
Standardize Column Years	▪ If a model has multiple tabs, make sure the years are in the same column on all tabs	Reduces the chances for error and/or linking into the wrong year in a schedule
Row Heights / Column Widths	▪ Use standard row heights and column widths throughout	Visually appealing
Font Type	▪ Arial (unless company font differs)	Most widely accepted and easiest to read
Background	▪ No color, no gridlines	“Cleaner” and easier to find numbers
Page Headers	▪ Top two rows on every tab includes appropriate titles	Audience knows what they are looking at on each page
Print-Ready	▪ Have model ready to print at all times	Enables audience to fully review analysis
Line Item Sum	▪ The sum row has an indentation of “1” with a border on top	Enables audience to clearly interpret information

Financial Modeling Best Practices (Cont'd)

Please see below for an example of a page from a financial model with best practices. As you can see, inputs are in blue font, links to other tabs are in green font, and calculations are in black font. All numbers are aligned on the right, enabling the reader to quickly analyze the figures. Currency symbols are at the top and bottom of each data set, and there are never two currency symbols in a row. All row heights and column widths are consistent throughout the spreadsheet, and the font type is Arial.

Allegiant Travel Financial Model		This document subject to the Conditions of Use							
Income Statement (\$ in Millions)		© 2013 Adventis CG Training LLC							
		Fiscal Year Ending Jan 31,							
		2016A	2017A	2018A	2019E	2020E	2021E	2022E	2023E
Income Statement									
Revenue		\$67,997	\$70,395	\$74,754	\$78,622	\$82,802	\$86,757	\$89,998	\$92,698
% Growth			3.5%	6.2%	5.2%	5.3%	4.8%	3.7%	3.0%
Cost of Sales		44,693	46,133	48,912	51,339	53,894	56,373	58,475	60,254
Gross Profit		\$23,304	\$24,262	\$25,842	\$27,283	\$28,908	\$30,384	\$31,523	\$32,444
% Margin		34.3%	34.5%	34.6%	34.7%	34.9%	35.0%	35.0%	35.0%
Operating Expenses		17,465	17,601	18,076	18,869	19,458	19,954	20,250	20,394
Operating Income		\$5,839	\$6,661	\$7,766	\$8,414	\$9,450	\$10,430	\$11,273	\$12,051
% Margin		8.6%	9.5%	10.4%	10.7%	11.4%	12.0%	12.5%	13.0%
Interest Expense		\$530	\$606	\$632	\$682	\$888	\$1,092	\$1,240	\$1,329
Other Expense		36	(13)	(87)	(25)	(17)	(17)	(8)	(5)
Total Other Expenses		\$566	\$593	\$545	\$657	\$871	\$1,075	\$1,232	\$1,324
Earnings before Taxes		\$5,273	\$6,068	\$7,221	\$7,757	\$8,578	\$9,355	\$10,041	\$10,727
Corporate Taxes		1,935	2,185	2,686	2,870	3,174	3,461	3,715	3,969
Net Earnings		\$3,338	\$3,883	\$4,535	\$4,887	\$5,404	\$5,893	\$6,326	\$6,758
% Margin		4.9%	5.5%	6.1%	6.2%	6.5%	6.8%	7.0%	7.3%
Metrics & Drivers									
Corporate Tax Rate		36.7%	36.0%	37.2%	37.0%	37.0%	37.0%	37.0%	37.0%
Operating Expenses as a % Revenue		25.7%	25.0%	24.2%	24.0%	23.5%	23.0%	22.5%	22.0%

12. Checklist for the Financial Modeling Exam

Once you've successfully participated in the first Financial Modeling Course and begin to practice on your own, the below tips will help you progress your skills and prepare for the FMC™ Level I Certification financial modeling exam.

No.	Hint	Examples
1	Always format top and bottom rows with currency; if there are two rows format the number under the sum line	▪ Cash from investing activities on cash flow statement ▪ Beginning and ending cash on cash flow
2	Always link beginning balance to the previous year's ending balance	▪ Equity schedule: beginning equity from the balance sheet ▪ Capital structure schedule: beginning cash from the balance sheet
3	Make sure all cells have the proper color formatting	▪ Red is only used on the "checks" on the balance sheet ▪ Blue is for inputs ▪ Green is for any formula that is linked to another sheet
4	Format key metrics appropriately	▪ Cash Ratio is a ratio and therefore formatted as a two decimal place number ▪ Debt-to-equity is a percentage ▪ Debt / EBITDA is a multiple and should have a "x"
5	The two currency symbols in a row rule is only broken in situations dealing with "per share" or share price	▪ Dividends per share, EPS, and share price all require two decimal places, as they are different denominations from most other figures in the financial model
6	Be conscious of positive or negative signage when calculating formulas	▪ Depreciation is a source of cash on the cash flow statement due to it being a non-cash expense ▪ Capital expenditures is a use of cash
7	Indent lines approximately to improve organization	▪ Items that are "floating" such as beginning and ending cash are both indented as well as every total line underneath an underline
8	Fully understand how the other debt issued affects other numbers depending on situation	▪ By adding other debt issued onto the other debt line of the debt schedule you allow your model to update if cash is not enough
9	Stress test the financial model before finishing and ensure key relationships hold	▪ Change the minimum cash so debt is needed to fund the business ▪ Change growth rates and margins

13. **FMC™ Level I Certification Scorecard**

In order to receive a FMC™ Level I Certification, trainees must pass all four categories listed out below.

Category	Commenatary
Completion time	The financial model must be completed in 90 minutes or less to pass this section. If the model is completed in more than 90 minutes, the candidate will not earn his/her FMC Level I Certification.
Formulas	Formulas in the model must be almost entirely correct to pass this section. If there is one critical formula error or several minor formula errors, the candidate will not earn his/her FMC Level I Certification.
Formatting	There must be 10 or less formatting mistakes in the financial model to pass this section. If there are 11 or more formatting mistakes, the candidate will not earn his/her FMC Level I Certification.
Multiple choice exam	A score of at least 90% on the FMC Level I Certification multiple choice exam must be achieved in order to pass this section. If the score is 89% or below, the candidate will not earn his/her FMC Level I Certification

14. Sample Questions for the FMC™ Level I Multiple Choice Exam

Below is a list of illustrative questions that are similar to what you will see on the 30-question FMC™ Level I multiple choice exam.

What is the definition of cost of goods sold?

- A. Direct cost of producing revenue
- B. The amount charged for the delivery of goods or services
- C. All other expenses required to run a business
- D. Expenses or income not related to the regular business of a company

Which of the following falls into cash flow from operating activities?

- A. Capital expenditures
- B. Working capital
- C. Debt issuances
- D. Dividends

Which of the following is not a source of cash?

- A. Depreciation
- B. Debt issuance
- C. Equity issuance
- D. Dividends

Answers: A, B, D

15. **Overview of Valuation Concepts**

“Value is what people are willing to pay for...”

– John Naisbitt

There are many different methods, formulas and other analytics that can be used to arrive at a valuation, however value is ultimately what the buyer pays.

The two primary types of valuation are relative and intrinsic. Relative valuation refers to methods that compare the price of a company to the market value of similar assets. Intrinsic valuation refers to the value of a company through fundamental analysis around its ability to generate cash flow.

Transaction Value vs. Purchase Price

We discussed the difference between Enterprise Value and Equity Value in the Financial Ratios section of this packet. In an M&A context, Enterprise Value is analogous to the Transaction Value and Equity Value is analogous to the Purchase Price. To illustrate how this works, assume there is a press release that says a company sold for \$100M. Furthermore, in the press release the company being bought had \$15M of debt and \$2M of cash. Regardless of how much debt or cash the acquired company had, it should still be worth \$100M. The shareholders of the acquired company would take the \$2M of cash to pay down remaining debt, leaving \$13M in debt. Then they would use the proceeds from the deal to pay down the remaining debt. The result would be \$87M (the \$100M transaction value less the \$13M in net debt). The \$87M represents the purchase price of the transaction (i.e. the “check” to shareholders of the acquired company).

Relative Valuation

Relative valuation refers to methods that compare the price of a company to the market value of similar assets. Two primary types of relative valuation are comparable company analysis and acquisition comparables analysis.

Comparable Companies Analysis

Comparable companies analyses (or public trading comparables analyses) are the most common type of relative valuation analyses utilized for companies. These methods allow investors to compare the valuation of similar companies by comparing similar ratios. The most common ratios are Enterprise Value / EBITDA, Enterprise Value / Revenue and Equity Value / Earnings (often calculated as Share Price / Earnings per Share). Assume a company has \$5M of EBITDA and the two publicly held companies that are most similar to the company trade in the market at 6.0x and 7.0X EBITDA. Using these multiples, you might conclude that the company's Enterprise Value should be worth \$30M - \$35M. Additionally, when a company trades at a multiple that is a premium or discount to the industry average, investors will often dig in to understand the rationale. Assume that a company trades at 7.0x EBITDA but the average of comparable companies is 9.0x; absent any significant business challenges, an investor may come to the conclusion that that company is undervalued and look to buy shares, realizing that the share price will appreciate if/when Wall Street begins to value the company in-line with its peers. Please see below for an example of a comparable company analysis.

Illustrative Comparable Companies Analysis								
	Enterprise Value	Equity Value	Enterprise Value /				Equity Value /	
			Revenue		EBITDA		Net Income	
			2019E	2020E	2019E	2020E	2019E	2020E
Company 1	\$2,100	\$2,300	0.9x	0.8x	4.4x	4.0x	13.7x	11.9x
Company 2	1,250	1,250	0.9x	0.8x	6.9x	5.8x	14.4x	11.7x
Company 3	1,200	1,550	1.0x	0.9x	4.5x	3.8x	18.5x	14.9x
Company 4	775	800	1.0x	0.8x	5.0x	4.5x	11.8x	9.4x
Company 5	775	875	1.0x	0.9x	6.9x	5.7x	12.2x	10.8x
Average			0.9x	0.8x	5.5x	4.8x	14.1x	11.7x
Median			1.0x	0.8x	5.0x	4.5x	13.7x	11.7x

Relative Valuation (Cont'd)

Acquisition Comparables Analysis

Acquisition comparables analysis (or transaction comparables analysis) is similar to comparable companies analysis, however acquisition comparables analysis represent comparable acquisitions that have taken place and been publicly announced. In a given industry, multiples for acquisition comparables are typically greater than multiples for comparable companies, because acquirers typically need to pay a premium to the current share price to gain control of the company. Please see below for an example of an acquisition comparables analysis.

Illustrative Acquisition Companies Analysis					
Date	Acquiror	Target	Transaction Value	Transaction Value / LTM	
				Revenue	EBITDA
6/1/2010	Company A	Company 1	\$750	0.9x	4.4x
9/24/2011	Company B	Company 2	200	0.7x	5.3x
3/16/2012	Company C	Company 3	1,500	1.2x	5.5x
1/18/2013	Company D	Company 4	150	0.9x	4.4x
9/29/2015	Company E	Company 5	300	1.1x	7.8x
Average				0.9x	5.5x
Median				0.9x	5.3x

Intrinsic Valuation

Intrinsic valuation refers to the value of a company through fundamental analysis without reference to its market value, instead looking to a company's ability to generate cash flow. The most commonly used method of intrinsic valuation is the discounted cash flow ("DCF") analysis. The discounted cash flow analysis discounts all projected future cash flow of a company to the present by using the concept of the time value of money.

DCF Cash Flow Projections

The cash flow metric that is typically used in a DCF analysis is unlevered free cash flow, which represents the cash flow available to all stakeholders in the business (i.e. it is not affected by capital structure and, accordingly, does not include interest expense). Tax-effected EBIT is used rather than net income because the valuation of a company should not be dependent on capital structure. Because interest expense is subtracted from EBIT to arrive at net income on the income statement, applying a corporate tax rate directly to EBIT without subtracting interest expense eliminates the impact of capital structure to cash flow. To arrive at unlevered free cash flow, adjustments for depreciation and amortization, change in working capital and capital expenditures are added to tax-effected EBIT. This cash flow is projected out in the projection period, which is typically 5 years. In certain circumstances, such as a start-up or high growth business, 10 years could be used. Whatever the time period used, the analyst should end the model with a financial year representative of a "steady state" to ensure the analysis does not over- or understate total valuation. Please see below for a representation of the line items that typically calculate unlevered free cash flow in a DCF analysis.

Illustrative DCF Cash Flow Projections					
	Fiscal Year Ending Dec 31,				
	2019E	2020E	2021E	2022E	2023E
Revenue	\$100	\$125	\$140	\$150	\$160
Less: Operating Expenses	89	112	125	133	141
EBIT	\$11	\$13	\$15	\$17	\$19
Taxes @ 35%	4	5	5	6	7
Tax-Effectd EBIT	\$7	\$8	\$10	\$11	\$12
Plus: Depreciation	\$5	\$6	\$7	\$8	\$9
Change in Working Capital	(1)	(1)	(2)	(2)	(3)
Less: Capital Expenditures	(8)	(8)	(9)	(9)	(9)
Unlevered Free Cash Flow	\$3	\$5	\$6	\$8	\$9

Intrinsic Valuation (Cont'd)

Present Value of the Projection Period

The first component of determining the present value of a company is to calculate the present value of each year's unlevered free cash flow. This is done by discounting each year's cash flow to the present value using a discount rate. The discount rate reflects the cost of capital from the perspective of the stakeholder who is analyzing a company. This is done using the formula depicted for present value at the end of this section.

Illustrative Present Value of the Projection Period					
	Fiscal Year Ending Dec 31,				
	2019E	2020E	2021E	2022E	2023E
Unlevered Free Cash Flow	\$3	\$5	\$6	\$8	\$9
Present Value @ a Discount Rate of 10%:	\$3	\$5	\$4	\$5	\$6
Present Value of Projection Period					23

Intrinsic Valuation (Cont'd)

Terminal Value

The second component of determining the present value of a company is to calculate the present value of all cash flow beyond the projection period. While it's difficult enough to project the cash flow over a 5- or 10-year projection period, it is far more difficult to project the cash flows over the entire life of a company. Therefore, investors use the terminal value approach to make assumptions about long-term growth and cash flow. The terminal value is determined using one of two methods: the perpetuity method and the EBITDA exit multiple method.

The perpetuity method assumes that the free cash flow in the last year of the projection period grows into perpetuity at an annual rate of growth (typically 2-3%). To calculate the terminal value under the perpetuity growth method, the Gordon Growth Model is utilized (please see this formula at the end of this section). This formula rests on the assumption that the cash flow of the last projected year will stabilize and continue at the same rate of growth forever. The perpetuity growth rate represents an average growth rate. Growth in some years will be higher or lower, but the expectation is that future growth will average the perpetuity growth assumption. In practice, you would typically expect to see growth decline as the company matures and for the perpetuity growth rate not to exceed the local rate of inflation as this would signify that the company will eventually grow to be larger than the entire domestic economy.

The EBITDA exit multiple assumes that the company is sold in the last year of the projection period at a multiple of EBITDA. Often times, investors will use one method and back into an implied value for the other method as a check (i.e. if a 12.0x EBITDA multiple exit implies a 5% perpetuity growth rate, the exit multiple may be considered unrealistic). The formulas used for these two methods are in the table at the end of this section.

Intrinsic Valuation (Cont'd)

These two methods of deriving terminal value are often used beside each other, however one or the other might be better suited for different situations. There are many different factors involved in determining if both or one of these methods is appropriate. Please see below for a depiction of how the terminal value is calculated using both methods.

Illustrative Terminal Value Calculations			
<u>Perpetuity Method</u>		<u>EBITDA Exit Multiple Method</u>	
Present Value of Projection Period	\$23	Present Value of Projection Period	\$23
2023E Free Cash Flow	\$9	2023E EBITDA	\$28
Perpetuity Growth Rate	3.5%	EBITDA Multiple	7.0x
Terminal Value	\$149	Terminal Value	\$196
Years	5.0	Years	5.0
Present Value of Terminal Value	\$92	Present Value of Terminal Value	\$122
Total Enterprise Value	\$115	Total Enterprise Value	\$145

Intrinsic Valuation (Cont'd)

Discount Rate

The discount rate used in a DCF analysis should be the cost of capital for the business being valued. The weighted average cost of capital ("WACC") is typically used to determine the discount rate or a range of discount rates to be used in a DCF analysis. Often companies add a premium to the WACC to determine a "hurdle rate" and be used as the discount rate.

Weighted Average Cost of Capital ("WACC")

Assume you want to purchase a company and you go to the local bank and borrow \$1,000 at a 6% interest rate. Every year you are going to have to write a check to the bank for \$60. Your cost of capital, therefore, is 6%. Now assume the company you purchase produces \$50 of cash flow (i.e. a 5% return). In this case, you lose money and the investment would not be worth it. Assume it returned 7%. You might argue 1% profit isn't worth it for the risk required to raise the capital and go purchase the company. However, you might add 3% to the 6% cost of debt and have a hurdle rate of 9%. Therefore, you require the \$1,000 acquisition to produce at least \$90 of cash flow to meet the hurdle rate. Now, assume instead of getting all \$1,000 from the bank, you get \$500 from the bank and \$500 from your uncle Leo who would like to see a 20% return on his capital. Now your WACC would be 13% and with the 3% addition you would require a 16% hurdle rate, or \$160 of cash flow. If the business being sold for \$1,000 were to give you their financial projections, you could put them into a DCF analysis and discount the cash flow projections at either 9% or 16%, depending on the capital structure of your capital. This is an oversimplified example that does not take tax deductions on interest expense into account, but it should serve as a means to explain the concept of the WACC.

Intrinsic Valuation (Cont'd)

The WACC is the minimum return that a company must earn on an existing asset base (investors and creditors) to satisfy all capital providers. WACC represents the blended cost to both debt holders and equity holders of a company based on the cost of debt and the cost of equity for that company. Please see below for an illustration of the WACC Formula.

WACC Formula	
$WACC = \frac{E}{V} * Re + \frac{D}{V} * Rd * (1 - Tc)$	
E = Market Value of the Company's Equity	Re = Cost of Equity
D = Market Value of the Company's Debt	Rd = Cost of Debt
V = E + D	Tc = Corporate Tax Rate
E/V = % of Financing that is Equity	
D/V = % of Financing that is Debt	

Intrinsic Valuation (Cont'd)

The Capital Assets Pricing Model ("CAPM")

The CAPM is used to calculate the cost of equity. It is the rate of return that equity owners expect and is equal to the risk free rate (typically the yield on a 10-year U.S. government bond) plus the company's beta (the measure of how volatile the stock is in relation to the stock market) multiplied by the market risk premium (the expected return of the market less the risk free rate). Please see below for an illustration of the WACC Formula.

CAPM Formula	
$Re = Rf + B \times (Rm - Rf)$	
Re = Cost of Equity	B = Beta
Rf = Risk Free Rate	Rm = Expected Return of the Market

Overview of Valuation Formulas

As you build the valuation schedules there will be a number of formulas you'll need to make calculations and build these concepts into your financial model. Please see below for an overview of these formulas:

Metric	Definition
Present Value ("PV")	$\text{Future Value} / (1 + \text{Discount Rate})^{\text{Term}}$
Terminal Value (Perpetuity Method)	$\frac{\text{Terminal Year FCF} \times (1 + \text{Perpetuity Growth Rate})}{(\text{Discount Rate} - \text{Perpetuity Growth Rate})}$
Terminal Value (Exit Multiple Method)	Terminal Year EBITDA x EBITDA Multiple
Present Value of Terminal Value	$\text{Terminal Value} / (1 + \text{Discount Rate})^{\text{Term}}$

16. Overview of Leveraged Buyout Concepts

A leveraged buyout analysis will tell you how much a private equity firm could afford to pay for a business. Private equity firms target a higher return profile than normal equity investors and therefore use higher leverage to generate higher returns. We teach our Leveraged Buyout Course as part of our FMC™ Program because it gives us an opportunity to teach key concepts such as transaction value, sources and uses of funds and investor returns.

Transaction Value

For publicly traded companies, the purchase price typically assumes a premium to where the stock price is currently trading to incent a change in ownership. The premium typically ranges between 20% and 40%. The transaction value is then determined by adding net debt currently on the balance sheet of the company being acquired. The transaction value represents the total value that must be financed to acquire and control the company.

Sources and Uses

The sources and uses reflect both the sources of capital to finance the acquisition and the uses of that capital. Sources typically include debt and equity used to finance the acquisition. Debt sources typically include bank debt, which is amortized over a period of time, and high yield debt is interest only with the ability to pay down principal based on cash flow. Uses typically include the transaction value, beginning cash balance and deal fees. Sources and uses are always equal to each other.

Financing Structure

The total amount and types of debt are dictated by lenders with any remaining capital necessary to finance the acquisition coming from the financial sponsor as their initial equity investment. The amount of debt will vary based on several factors, including the industry the company operates in and the predictability of cash flow. Debt is typically split across two tranches: senior debt and subordinated debt. Senior debt has a first claim in bankruptcy and is typically amortized (paid back) over the life of the loan and therefore has a lower interest rate. Subordinated debt typically requires interest payments only with the ability to voluntarily pay down principal with any excess cash flow. Subordinated debt also has higher interest rates due to the additional default risk these debt investors assume. Debt / EBITDA ratios are typically 2.0x – 4.0x and total Debt / EBITDA ratios are typically 4.0x – 6.0x. The financial sponsor's equity investment then makes up any remaining capital required to finance the transaction value and typically represents between 20-40% of total sources.

Investor Returns

There are several measures of returns that financial sponsors analyze as they structure a transaction (and decide what price they can pay and how much equity to invest) and consider their return objectives. The most commonly used metrics include the cash-on-cash return and the internal rate of return ("IRR"). Both measures take into account the relationship of capital invested by the financial sponsor versus capital returned over the life of the investment. The IRR, unlike cash-on-cash returns, also takes into account the time value of money and produces the annualized rate of return generated by the investment. Target IRRs typically range from 20% to 30%. Please see the table below for formulas for these investor return metrics.

Metric	Definition	Formula	Example
Internal Rate of Return (aka IRR)	The annualized effective compounded return rate that makes the net present value equal to zero	$(\text{Cash Returned to Sponsor} / \text{Initial Equity Invested})^{(1 / \text{Investment Time Horizon})} - 1$	If an investor receives a 15% IRR over the life of an investment it means that his investment increased on average 15% per year
Cash-on-Cash Multiple	How much an investor receives in proceeds upon exiting the investment compared to its initial investment; it is not dependent on when the exit actually occurs	$\text{Cash Returned to Sponsor} / \text{Initial Equity Invested}$	If an investor gets a cash-on-cash multiple of 2.5x, it means they received \$2.50 for every \$1.00 they invested

17. Sample Questions for the FMC™ Level II Multiple Choice Exam

Below is a list of illustrative questions that are similar to what you will see on the 25-question FMC™ Level I multiple choice exam.

Which of the following belongs with Enterprise Value in a multiple?

- A. Net Earnings
- B. Taxes
- C. EBITDA
- D. None of these belong with Enterprise Value

Which of the following statements is true?

- A. Transaction value is synonymous with equity value
- B. Transaction value – net debt = enterprise value
- C. Equity value + net debt = enterprise value
- D. All of these statements are false

If a company is announced to be sold for a transaction value of \$15 million and it has \$2 million of debt and \$1.5 million of cash, what is the purchase price of the company?

- A. \$17 million
- B. \$15 million
- C. \$14.5 million
- D. \$15.5 million
- E. \$13 million

Answers: C, C, C,

18. Frequently Asked Questions - Interviews

This section doesn't have anything that isn't in the body of this course packet. It simply articulates the information in a question and answer format to help you prepare for interviews. It is broken up into three sections with selected questions on financial statements, valuation, and leveraged buyouts.

Financial Statement Questions

Walk me through the three financial statements.

The three financial statements are the income statement, balance sheet, and cash flow statement. The income statement gives an organization's revenue, expenses and resulting net income over a certain period. The balance sheet shows an organization's financial position at a point in time and discloses the resources an organization controls (assets), as well as the claims against those resources (liabilities and equity). The cash flow statement begins with net income and shows the change in cash for operating, investment, and financing activities of an organization. The cash flow statement reconciles net income to change in cash. The bottom line of the cash flow statement is cash and reconciles with cash on the balance sheet.

How do the three financial statements link together?

Net income from the income statement flows into shareholders' equity on the balance sheet and appears as the first line item on the cash flow statement. Changes to working capital items on the balance sheet, debt, and other balance sheet items appear as sources or uses of cash on the cash flow statement. Ending cash on the cash flow statement links into the balance sheet. Depreciation and amortization is an expense on the income statement (it is typically included within COGS and / or other expenses) that reduces the value of Property, Plant & Equipment ("PP&E") on the balance sheet and appears on the cash flow statement as a source of cash. Conversely, capital expenditures ("CapEx") appears as a use of cash on the cash flow statement and reduces PP&E on the balance sheet. Any changes in debt impact debt on the balance sheet and appear as a source or use of cash on the cash flow statement. Dividends and share repurchases reduce shareholders' equity and appear as a use of cash on the cash flow statement. Interest expense on the income statement is calculated based on the debt outstanding on the balance sheet.

Financial Statement Questions (Cont'd)

Walk me through how depreciation of \$100 would affect the three financial statements.

Expenses on the income statement would increase by \$100 and, assuming a 40% tax rate, taxes would be reduced by \$40 which would reduce net income by \$60. On the cash flow statement, depreciation of \$100 would be added back because it is a non-cash expense. Resulting cash flow from operations would be increased by \$40. On asset side of the balance sheet, PP&E would be reduced by \$100, cash would be increased by \$40 and total assets would therefore be decreased by \$60. On the liabilities side of the balance sheet, shareholders' equity would be decreased by \$60.

Valuation Questions

What are the three major valuation methodologies?

The three major valuation methodologies are comparable companies analysis, precedent transaction analysis, and discounted cash flow (“DCF”) analysis. Comparable companies analysis and precedent transaction analysis are both examples of relative valuation, which compares the price of a company to the market value of similar publicly traded companies. DCF analysis is an example of intrinsic valuation, which refers to the value of a company through fundamental analysis and values the firm on the basis of its cash flow generating potential without reference to its market value.

What are the most common multiples used in valuation?

The most common multiples are Enterprise Value (“EV”) / revenue, EV / EBITDA and Price / Earnings (“P/E”).

Why can't you use Equity Value / Enterprise Value, rather than Enterprise Value / EBITDA?

It's important that both numbers are philosophically similar. Enterprise Value Shows the sum of all ownership interests in the company and is independent of capital structure. Similarly, EBITDA is available to all shareholders and is independent of capital structure. Because equity value is the value of a company available to shareholders only, comparing equity value to EBITDA would be like comparing apples to oranges.

Walk me through a Discounted Cash Flow (“DCF”) analysis.

A DCF analysis discounts all projected future cash flow of a company to the present by using the concept of the time value of money. In order to build a DCF analysis, you would project free cash flow for a period of time, typically five years. This is done by tax effecting EBIT to get tax-affected EBIT and adjusting for D&A, CapEx, and change in working capital. The result is unlevered free cash flow, the cash flow available to all stakeholders.

You would then discount all unlevered free cash flow in the projection period back to the present value using a discount rate. You would determine the discount rate using the Weight Average Cost of Capital, or WACC. Then, you would value all cash flow beyond the projection period by determining a terminal value. There are two methods for calculating the terminal value, the Gordon Growth (aka perpetuity growth method) or the exit multiple method. The Gordon Growth method assumes that the last year of unlevered free cash flow from the projection period grows at a rate into perpetuity. This growth rate should be modest, for example, average long-term expected GDP growth or inflation. The exit multiple method assumes the company is bought in the last year of the projection period for a multiple of EBITDA. Once the terminal value is calculated, you would then discount the terminal value to the present value. .

Valuation Questions (Cont'd)

What is the difference between enterprise value and equity value?

Enterprise Value represents the value of the operations of a company attributable to all providers of capital (debt and equity). Equity Value is one of the components of enterprise value and represents only the proportion of value attributable to shareholders.

Leveraged Buyout Questions

Walk me through a basic leveraged buyout analysis.

First, we need to make some transaction assumptions. We would need to determine the purchase price by applying a premium to where the stock price is currently trading. Then we would add net debt to arrive at the transaction value, the total worth of the firm. We would then build a sources and uses table. Uses of capital reflect the amount of money required for the transaction and would include the transaction value, desired beginning cash balance, and transaction expenses. Sources of capital reflect where the capital comes from and would include debt and equity. Sources will always equal uses.

The next step would be to construct “pro forma” financial statements. Key balance sheet items would include debt, equity, and cash; key income statement items would include revenue, EBIT, interest expense, taxes, and resulting net income. Key cash flow items would include depreciation and amortization, change in working capital, capital expenditures, and debt pay down. Interest expense would be calculated based on debt balances. Once the financial statement projections are created, we would then make assumptions about the private equity firm’s exit from its investment, calculated resulting enterprise and equity values and determine the resulting investor returns.

What makes an ideal leveraged buyout candidate?

Ideal candidates would have steady and predictable operating cash flow, minimal capital requirements, the ability to expand margins, low working capital requirements, a strong base of assets for collateral, and a strong management team.